1. Smart Analytics, Machine Learning, and AI on Google Cloud

Description:

Incorporating machine learning into data pipelines increases the ability to extract insights from data. This course covers ways machine learning can be included in data pipelines on Google Cloud. For little to no customization, this course covers AutoML. For more tailored machine learning capabilities, this course introduces Notebooks and BigQuery machine learning (BigQuery ML). Also, this course covers how to productionalize machine learning solutions by using Vertex AI.

1.1. Introduction

In this module, we introduce the course and agenda

1.1.1. Video - Course Introduction

- YouTube: Course Introduction

1.2. Introduction to Analytics and AI

This modules talks about ML options on Google Cloud

1.2.1. Video - Module Introduction

- YouTube: Module Introduction

1.2.2. Video - What is AI?

- YouTube: What is AI?

1.2.3. Video - From ad-hoc data analysis to data-driven decisions

- YouTube: From ad-hoc data analysis to data-driven decisions

1.2.4. Video - Options for ML models on Google Cloud

- YouTube: Options for ML models on Google Cloud

1.2.5. Quiz - Introduction to Analytics and AI

1.2.5.1. Quiz 1.

> **Important**
>
> **What is the difference between AI and ML?**
>
> - [ ] AI is ML but without mathematics
> - [x] AI is a discipline while ML is a toolset
> - [ ] AI and ML are the same
> - [ ] AI concentrates on algorithms while ML is about theory

1.2.5.2. Quiz 2.

> **Important**
>
> **What is the primary impact of ML?**
>
> - [x] Provides insights that were not previously possible
> - [ ] Cost savings
> - [ ] It allows businesses to be more accurate in their predictions
> - [ ] It allows business operations to scale

1.3. Prebuilt ML Model APIs for Unstructured Data

This module focuses on using pre-built ML APIs on your unstructured data

1.3.1. Video - Module introduction

- YouTube: Module introduction

1.3.2. Video - Unstructured data is hard

- YouTube: Unstructured data is hard

1.3.3. Video - ML APIs for enriching data

- YouTube: ML APIs for enriching data

1.3.4. Video - Lab Intro: Using the Natural Language API to Classify Unstructured Text

- YouTube: Lab Intro: Using the Natural Language API to Classify Unstructured Text

1.3.5. Lab - Using the Natural Language API to classify unstructured text

In this lab you'll learn how to classify text into categories using the Natural Language API

- [x] Using the Natural Language API to classify unstructured text

1.3.6. Quiz - Prebuilt ML Model APIs for Unstructured Data

1.3.6.1. Quiz 1.

> **Important**
>
> **True or False? Most business data is unstructured data, and mainly text.**
>
> - [x] True
> - [ ] False

1.3.6.2. Quiz 2.

> **Important**
>
> **Google Cloud's pretrained model APIs use:**

> - [ ] Your models and Google's data
> - [ ] Your models and your data
> - [ ] Google's models and Google's data
> - ✓ Google's models and your data

1.4. Big Data Analytics with Notebooks

This module covers how to use Notebooks

1.4.1. Video - Module introduction

- YouTube: Module introduction

1.4.2. Video - What's a Notebook?

- YouTube: What's a Notebook?

1.4.3. Video - BigQuery magic and ties to Pandas

- YouTube: BigQuery magic and ties to Pandas

1.4.4. Video - Lab Intro: BigQuery in JupyterLab on Vertex AI

- YouTube: Lab Intro: BigQuery in JupyterLab on Vertex AI

1.4.5. Lab - BigQuery in JupyterLab on Agent Platform 2.5

The purpose of this lab is to show learners how to instantiate a Jupyter notebook running on Agent Platform.

- ✓ BigQuery in JupyterLab on Agent Platform 2.5

1.4.6. Quiz - Big Data Analytics with Notebooks

1.4.6.1. Quiz 1.

> **Important**
> **True or False? Notebooks contain a magic function to execute BigQuery**
>
> - [x] True
> - [ ] False

1.4.6.2. Quiz 2.

> **Important**
> **Select the statement that does not apply to Notebooks.**
>
> - [x] It's up to you to install the latest ML libraries on the notebooks
> - [ ] You can easily change hardware including adding and removing GPUs
> - [ ] They use the latest open-source version of JupyterLab
> - [ ] Notebook instances are standard Compute Engine instances that live in your projects

1.5. Production ML Pipelines

This module covers building custom ML models and introduces Vertex AI and TensorFlow Hub

1.5.1. Video - Module introduction

- YouTube: Module introduction

1.5.2. Video - Ways to do ML on Google Cloud

- YouTube: Ways to do ML on Google Cloud

1.5.3. Video - Vertex AI Pipelines

- YouTube: Vertex AI Pipelines

1.5.4. Video - TensorFlow Hub

- YouTube: TensorFlow Hub

1.5.5. Video - Lab Intro: Running Pipelines on Vertex AI

- YouTube: Lab Intro: Running Pipelines on Vertex AI

1.5.6. Lab - Running Pipelines on Agent Platform 2.5

In this lab, you learn how to utilize Agent Platform Pipelines to execute a simple Kubeflow Pipeline SDK derived ML Pipeline.

- [ ] Running Pipelines on Agent Platform 2.5

1.5.7. Video - Summary

- YouTube: Summary

1.5.8. Quiz - Production ML Pipelines

1.5.8.1. Quiz 1.

> **Important**
> **Which technology was developed as a solution to run Kubernetes clusters and pods behind the scenes to support deploying pipelines?**
>
> - [ ] Kubeflow
> - [ ] Cloud Orchestrator
> - [ ] Cloud Composer
> - [x] Vertex Pipelines

1.5.8.2. Quiz 2.

> **Important**
> **TensorFlow Hub has templates for which of the following?**
>
> - [ ] Jupyter notebooks
> - [ ] Kubeflow pipelines and components
> - [x] All other answers are correct
> - [ ] Trained models

1.6. Custom Model Building with SQL in BigQuery ML

This module covers BigQuery ML

1.6.1. Video - Module introduction

- YouTube: Module introduction

1.6.2. Video - BigQuery ML for Quick Model Building

- YouTube: BigQuery ML for Quick Model Building

1.6.3. Video - Supported models

- YouTube: Supported models

1.6.4. Video - Lab Intro: Predict Bike Trip Duration with a Regression Model in BigQuery ML

- YouTube: Lab Intro: Predict Bike Trip Duration with a Regression Model in BigQuery ML

1.6.5. Lab - Predict Bike Trip Duration with a Regression Model in BQML 2.5

In this lab you will use the London bicycles dataset to build a regression model in BQML to predict trip duration.

- [ ] Predict Bike Trip Duration with a Regression Model in BQML 2.5

1.6.6. Video - Lab Intro: Movie Recommendations in BigQuery ML

- YouTube: Lab Intro: Movie Recommendations in BigQuery ML

1.6.7. Lab - Movie Recommendations in BigQuery ML 2.5

In this lab you'll use the MovieLens dataset to build a collaborative filtering model and use it to make predictions.

- [ ] Movie Recommendations in BigQuery ML 2.5

1.6.8. Video - Summary

- YouTube: Summary

1.6.9. Quiz - Custom Model Building with SQL in BigQuery ML

1.6.9.1. Quiz 1.

> **Important**
>
> **True or False? You can train and evaluate machine learning models directly in BigQuery.**
>
> - [x] True
> - [ ] False

1.6.9.2. Quiz 2.

> **Important**
>
> **BigQuery ML has support for which of the following modeling tasks:**
>
> - [x] Regression
> - [ ] Computer vision
> - [x] Clustering
> - [x] Classification

1.7. Custom Model Building with Vertex AI AutoML

Custom model building with Vertex AI AutoML

1.7.1. Video - Module introduction

- YouTube: Module introduction

1.7.2. Video - Why AutoML?

- YouTube: Why AutoML?

1.7.3. Video - AutoML Vision

- YouTube: AutoML Vision

1.7.4. Video - AutoML Natural Language Processing

- YouTube: AutoML Natural Language Processing

1.7.5. Video - AutoML Tables

- YouTube: AutoML Tables

1.7.6. Video - Summary

- YouTube: Summary

1.7.7. Quiz - Custom Model Building with AutoML

1.7.7.1. Quiz 1.

> **Important**
> **AutoML makes use of which of the following:**
>
> - [x] Google's models and your data
> - [ ] Google's models and Google's data
> - [ ] Your models and Google's data
> - [ ] Your models and your data

1.7.7.2. Quiz 2.

> **Important**
> **Which of the following are valid techniques for improving AutoML Vision and AutoML Natural Language models?**

> - [x] Increase the amount of training data
> - [ ] Increase the number of labels
> - [x] Increase the diversity and complexity of data
> - [x] Ensure consistent labeling

1.8. Summary

This module recaps the topics covered in the course

1.8.1. Video - Course Summary

- YouTube: Course Summary

1.9. Course Resources

PDF links to all modules

1.9.1. Document - Smart Analytics, Machine Learning, and AI on Google Cloud

- T-SMARTA-I-8-l1-file-en-34-1.en.pdf

1.10. Your Next Steps

1.10.1. Credential - Claim credential

1.10.2. Survey - Course Survey